ReefPulse

Ocean Safety, Hazard Forecasting, and Marine Life Discovery for the California Current

Project Proposal

DataHacks 2026

Theme: Environment, Climate & Energy Sciences

Tracks: Machine Learning / AI and Cloud

Primary Datasets: CalCOFI, Spray Glider, CCE Mooring, iNaturalist (Scripps Institution of Oceanography)

1. Executive Summary

ReefPulse is a unified ocean intelligence platform built for the people who actually live and play on the California coast: surfers, swimmers, divers, and fishers, together with the coastal managers and public health officials who keep them safe. The platform forecasts ocean conditions up to fourteen days in advance, flags harmful algal blooms and unsafe swimming conditions before they become dangerous, and tells users what marine life they are likely to encounter on any given day at any given beach.

Our system combines four machine learning models in a single layered pipeline: a state forecaster that predicts temperature, salinity, dissolved oxygen, and chlorophyll-a; an anomaly detector that flags emerging marine heatwaves and acidification events; a harmful algal bloom classifier that delivers probabilistic bloom forecasts seven to fourteen days ahead; and a marine life predictor that estimates encounter probabilities for different fish and wildlife species at each location. A Surf Safety Index fuses these outputs with live wave, wind, and tide data into a single at-a-glance score for ocean users. A conversational Gemini-powered agent sits on top of the stack, letting anyone ask natural-language questions such as "is it safe to swim at La Jolla Cove tomorrow, and what might I see?".

ReefPulse runs on a production-grade AWS architecture spanning Kinesis, Lambda, SageMaker, EventBridge, and SNS. Events detected in the data pipeline trigger automated alerts to registered stakeholders within seconds. This proposal covers the project motivation, technical architecture, team plan, hour-by-hour timeline, risk management, and submission strategy across the nine challenges we are targeting.

2. Problem Statement and Target Users

2.1 The Core Problems

Millions of Californians rely on the ocean for recreation, income, and food, yet the information they need to stay safe and make smart decisions is fragmented across government portals, scientific reports, and word of mouth. ReefPulse addresses three concrete problems at once:

- **Unsafe swimming and surfing conditions.** Surfers and swimmers often cannot tell whether today's conditions are hazardous until they are already in the water. Bacterial contamination, rip currents, marine heatwaves, and poor water clarity all go underreported at the beach level.
- **Harmful algal bloom (HAB) risk.** HABs cause beach closures, shellfish poisoning, and ecosystem damage. Current advisories lag the actual bloom by days and rarely provide forward-looking forecasts.
- **Marine ecosystem damage.** Warming waters and shifting currents are rearranging where fish, mammals, and kelp live. Coastal managers need early warning to issue closures and manage resources, and recreational users want to understand what they are seeing change around them.

2.2 Target Users

ReefPulse is built for two tiers of users. The consumer tier is where the viral product lives; the operator tier is where institutional value is captured.

Consumer Tier (recreational ocean users)

- **Surfers.** Get a Surf Safety Index that combines water quality, HAB risk, wave conditions, wind, and tide into one score per break.
- **Swimmers and beachgoers.** See whether today's beach is safe, and why it would not be.
- **Divers and snorkelers.** Get visibility forecasts, current warnings, and a preview of the species likely to be seen at each dive site.
- **Recreational fishers.** See expected species hotspots based on current oceanographic conditions, species-environment models, and seasonal patterns.

Operator Tier (decision-makers)

- **Coastal managers.** Receive automated alerts when HAB probability, anomaly scores, or water-quality metrics cross closure thresholds.
- **Public health officials.** Monitor the same signals at a county or jurisdictional level to support advisories.
- **Aquaculture operators.** Get early warnings for conditions that could wipe out shellfish harvests.
- **Researchers.** Use the platform as a queryable interface over the fused dataset.

2.3 Why Now

The California Current is one of the four most productive ocean ecosystems on Earth. It is also increasingly unstable. Marine heatwaves like the 2013 to 2016 "Blob" devastated fisheries and caused mass seabird die-offs. Harmful algal blooms, driven by shifting nutrient regimes and warming waters, shut down shellfish harvesting for weeks at a time and occasionally cause human illness through domoic acid contamination. Ocean acidification is quietly eroding the foundations of the shellfish industry, with hatchery collapses already documented in the Pacific Northwest.

Despite the stakes, the tools available to most stakeholders are fragmented. Commercial fishing fleets rely on weather forecasts and word of mouth. Aquaculture operators receive sporadic advisories from state agencies, often after damage has already begun. Public health officials depend on reactive water sampling that takes days to return results. Recreational users have almost nothing that gives them a forward-looking view of safety and the marine life they may encounter.

Scripps Institution of Oceanography maintains some of the world's richest ocean datasets, including the CalCOFI time series that has operated continuously since 1949. Combined with modern real-time platforms like Spray gliders, the CCE mooring network, and crowdsourced species sightings from iNaturalist, these data provide an extraordinary foundation for building an operational forecasting and discovery system. Recent advances in time-series transformers, species distribution models, vision models, and large language models make it possible to fuse these heterogeneous data streams into a single coherent product for the first time.

3. Project Objectives

At the highest level, ReefPulse delivers three core functions to end users:

- **Ocean recreation safety in real time.** A single score per beach, surf break, or dive site indicating whether conditions are safe.
- **Environmental risk forecasting.** Fourteen-day forecasts of temperature, oxygen, chlorophyll, and HAB probability, with proactive alerts when thresholds are crossed.
- **Ecosystem-aware recommendations.** A marine life discovery feed telling users what they are likely to see based on conditions, season, and location.

By the conclusion of DataHacks 2026, the team will deliver:

1. A trained state-forecasting model that predicts temperature, salinity, dissolved oxygen, and chlorophyll-a across the California Current fourteen days ahead, with calibrated uncertainty bands, using post-1993 CalCOFI CTD data.
2. A variational-autoencoder-based anomaly detector that flags unusual ocean states in near real time, capable of identifying precursors to marine heatwaves.
3. A harmful algal bloom classifier combining satellite chlorophyll imagery with mooring nutrient time-series, delivering probabilistic bloom forecasts by coastal location.
4. A marine life predictor that estimates which species are most likely to be encountered at a given location and time, trained on CalCOFI larval fish records and iNaturalist sightings.
5. A Surf Safety Index that fuses model outputs with live wave, wind, and tide data into a single consumer-friendly score.
6. A conversational Gemini-powered agent that lets non-technical users query the system in plain English.
7. A deployed AWS architecture with event-driven alerting, serverless inference, and automated retraining.
8. A public-facing React dashboard that surfaces conditions, forecasts, alerts, and species discovery on an interactive map.
9. A polished three-minute pitch and demonstration targeting nine distinct prize categories.

4. Datasets

Primary datasets are drawn from Scripps Institution of Oceanography, making the project eligible for the $1,500 Scripps Challenge. iNaturalist data (referenced in the DataHacks dataset document under the species category) powers the marine life discovery feature. External supplements come from NOAA CoastWatch for satellite imagery and NOAA for wave and surf conditions.

Source	What It Provides	Update Rate	Format / Size
CalCOFI Bottle Database	75 years of historical T, S, O2, chlorophyll-a, nutrients, plus larval fish abundance across 250+ species	Quarterly	CSV / SQL, ~1M rows
Spray Glider Data	Real-time subsurface profiles along glider trajectories	~Hourly	NetCDF, ~GB scale
CCE Mooring Data	Fixed-point real-time telemetry in the CA Current Ecosystem	Sub-hourly	Streaming / CSV
MBARI Ocean Data	Central California coverage including biological sensors	Variable	Mixed repositories
iNaturalist (Scripps-listed)	Crowdsourced marine species sightings with geo-tags, timestamps, and photos	Daily	CSV export via API
NOAA CoastWatch (external)	Satellite-derived SST and chlorophyll-a imagery for bloom detection	Daily	Raster / NetCDF
NOAA Surf / Wave Data (external)	Wave height, period, wind, and tide forecasts for surf safety	Hourly	API (JSON)

To keep training tractable within the hackathon window, we will restrict the CalCOFI historical set to the post-1993 CTD-Rosette era. This avoids instrumentation-driven distribution shift between legacy bottle samples and modern continuous profiling. For iNaturalist we will filter to California coastal waters within the last ten years to keep the species occurrence set focused.

5. Technical Architecture

5.1 System Diagram

The architecture below shows the end-to-end flow from raw data sources through AWS ingestion, into the four-model ML layer, into the fusion and serving layer, and out to the two user tiers.

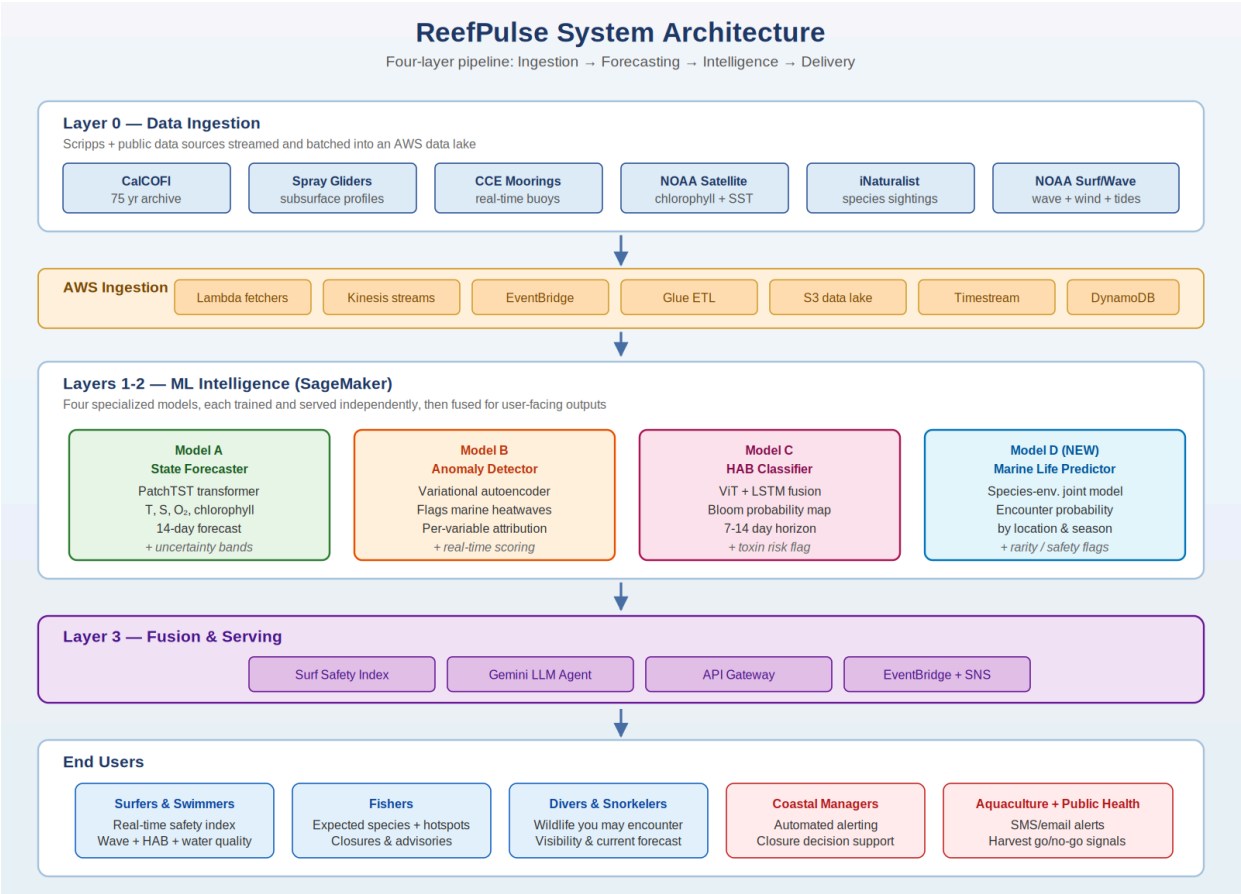

Figure 1. ReefPulse end-to-end architecture. Data flows from left-to-right at the source layer, top-to-bottom through AWS ingestion, into four independently-trained ML models, converges in the fusion layer (Surf Safety Index, Gemini agent, API, alerting), and is delivered to consumer and operator user tiers.

5.2 System Layers

ReefPulse is organized into four logical layers, each a separate concern that feeds the next:

- **Layer 0 - Data Ingestion:** Pulls CalCOFI archives, glider telemetry, mooring streams, iNaturalist species sightings, and satellite imagery into an S3 data lake.

- **Layer 1 - State Forecasting:** The PatchTST-based model predicts the full ocean state (temperature, salinity, oxygen, chlorophyll) fourteen days ahead.
- **Layer 2 - Intelligence:** Anomaly detection, HAB classification, and marine life prediction run in parallel, consuming the same underlying features.
- **Layer 3 - Fusion and Serving:** The Surf Safety Index combines outputs into a single consumer score; the Gemini agent provides a natural-language interface; EventBridge and SNS handle alerting; API Gateway serves the React dashboard.

5.3 Machine Learning Stack

Model A: State Forecaster

- **Architecture:** PatchTST (patch-based time-series transformer).
- **Input:** Last 30 days of mooring and glider telemetry, plus historical CalCOFI climatology used as priors.
- **Output:** 14-day forecast of temperature, salinity, dissolved oxygen, and chlorophyll-a at each station, with uncertainty bands from Monte Carlo dropout.
- **Training:** 75 years of CalCOFI quarterly data interpolated onto a regular grid for pretraining, followed by fine-tuning on real-time streams.

Model B: Anomaly Detector

- **Architecture:** Variational autoencoder trained on encoded representations of normal ocean states.
- **Input:** Current oceanographic state vector.
- **Output:** Anomaly score with per-variable attribution so operators can see which signals are driving the alert.
- **Use:** Flags marine heatwaves, acidification events, and unusual stratification in near real time.

Model C: HAB Classifier

- **Architecture:** A Vision Transformer on satellite chlorophyll imagery, fused via cross-attention with an LSTM on mooring nutrient time-series.
- **Input:** Satellite chlorophyll-a patches, Model A's forecast, and nutrient time-series.
- **Output:** Probability map of bloom occurrence at coastal locations 7 to 14 days ahead, plus a toxin risk flag for HAB species that produce domoic acid.
- **Training Target:** Historical HAB events curated from California Department of Public Health reports, supplemented by chlorophyll threshold crossings as proxy labels.

Model D: Marine Life Predictor (new)

This is the core of the species discovery feature. Given a location, date, and current ocean conditions, the model estimates the encounter probability for a curated set of marine species (fish, mammals, birds, invertebrates, kelp).

- **Architecture:** A species distribution model combining gradient-boosted trees (LightGBM) for tabular features with learned embeddings for species and season. For

richer deployments this can be extended to a graph neural network linking species that co-occur.

- **Input:** Location (lat/lon), date, sea-surface temperature, chlorophyll-a, dissolved oxygen, season, and nearby historical sighting density.
- **Output:** A ranked list of species with encounter probability, rarity flag, and safety annotation (e.g., "white shark - rare but present", "garibaldi - very likely").
- **Training data:** CalCOFI larval fish abundance records for pelagic species, fused with iNaturalist geo-tagged sightings for coastal and charismatic megafauna. Training labels are presence points; we use pseudo-absence sampling for negatives.
- **Why this matters:** It turns ReefPulse from a warning tool into a discovery tool. Users check the app to see conditions - and stay for the wildlife preview. This is the viral hook that drives the Most Viral Idea challenge submission.

The Surf Safety Index (fusion layer)

A rule-based scoring function calibrated against historical beach closures, combining Model A (oxygen, chlorophyll), Model B (anomaly score), Model C (HAB probability), and live NOAA wave/wind/tide data. The output is a 0-100 score with a traffic-light color and a plain-English explanation ("Moderate risk - elevated chlorophyll and 6ft swell. Experienced surfers only."). Conservative thresholds err on the side of caution, and the UI always links to official closure notices when they exist.

LLM Agent (Gemini)

A conversational layer built on the Gemini API lets users ask questions in plain English. The agent calls the model endpoints, consults a small RAG store of NOAA advisories and local knowledge, and returns answers with confidence levels and data sources cited. This directly targets the Most Innovative Build With AI challenge and the MLH Best Use of Gemini API challenge.

5.4 Cloud Architecture (AWS)

Our cloud stack is deployed entirely on AWS, maximizing alignment with the Cloud Track and the Best Use of AWS Services challenge. The design prioritizes event-driven orchestration, serverless compute where possible, and a full MLOps story.

Ingestion and Storage

- **AWS Lambda:** Scheduled fetchers pull Scripps APIs and iNaturalist exports into S3.
- **Amazon Kinesis Data Streams:** Ingests glider and mooring telemetry.
- **Amazon S3:** Raw zone plus processed Parquet zone for the data lake.
- **Amazon EventBridge:** Schedules satellite image pulls from NOAA CoastWatch.

Processing

- **AWS Glue:** ETL from raw into analysis-ready Parquet.
- **Amazon DynamoDB:** Low-latency cache of the latest state and species encounter tables for dashboard reads.
- **Amazon Timestream:** Native time-series store for telemetry queries.

ML Training and Serving

- **Amazon SageMaker Training Jobs:** Nightly retraining runs for all four models.
- **Amazon SageMaker Experiments:** Tracks model versions.
- **Amazon SageMaker Endpoints:** Hosts the four models behind HTTPS.
- **AWS Lambda + Amazon API Gateway:** Thin serving layer that orchestrates calls across models and computes the Surf Safety Index.

Alerting and Frontend

- **Amazon EventBridge Rules:** Fire when anomaly scores, HAB probabilities, or surf safety degrade past thresholds.
- **Amazon SNS:** Sends SMS and email alerts to registered users and operators.
- **Amazon CloudFront + S3:** Host and cache the React dashboard.
- **AWS Amplify:** Authentication for the user and operator portals.
- **Amazon CloudWatch:** Dashboards and alarms we can point judges at.

6. User Experience

The dashboard is organized around a map of the California coast. Users select a location (a beach, surf break, or dive site) and see three stacked panels:

- **Today panel.** Surf Safety Index with color and plain-English explanation, current water temperature, wave height, visibility, and any active advisories.
- **Forecast panel.** 14-day outlook with sliders, showing forecasted temperature, chlorophyll, and HAB probability. Anomaly flags appear as icons on the timeline.
- **Discovery panel.** A scrollable feed of the top 8-12 species a user is likely to encounter at this location today, with photos, encounter probability, and rarity/safety tags. Users can tap through to learn more, and the app links each entry back to iNaturalist for citizen-science reporting.

A persistent "Ask ReefPulse" button opens the Gemini agent for free-form questions like "I'm diving at La Jolla tomorrow morning - what should I know?" The agent grounds its answer in the model outputs for that location and time, with all sources cited.

7. Team Plan

The project is scoped for a four-person team, with each member owning one of four vertical slices. Where a teammate's skills overlap, we will pair on the most critical path of the week.

Role	Ownership	Core Skills	Key Deliverables
ML Lead	Model A (State Forecaster), Model B (Anomaly Detector), Model D (Marine Life Predictor)	PyTorch, time-series modeling, species distribution modeling, evaluation methodology	Trained models, evaluation notebooks, model cards
Data & HAB Lead	Data pipelines, iNaturalist ingestion, Model C (HAB Classifier)	Python data wrangling, computer vision, domain research	Cleaned datasets, species occurrence dataset, satellite pipeline, HAB classifier
Cloud Engineer	Full AWS architecture, CI/CD, infrastructure as code	AWS services (SageMaker, Lambda, Kinesis), Docker, Terraform/CDK	Deployed stack, serving endpoints, alert system
Product & Frontend Lead	React dashboard, surfer-friendly UI, Gemini LLM agent, pitch and demo	React, Tailwind, mapping libraries, prompt engineering, storytelling	Dashboard, species discovery feed, chatbot, 3-minute demo and slide deck

If our team is uneven in experience, the ML Lead and Data & HAB Lead should pair on modeling during the first eight hours, and the Cloud Engineer and Product Lead should pair on demo integration during the final eight hours.

8. Timeline

8.1 Pre-Hackathon Preparation (the week before)

- Download CalCOFI Bottle Database CSV and complete initial exploratory analysis.
- Export the California-coastal iNaturalist species set; build an initial occurrence table.
- Request AWS credits and obtain Gemini API keys.
- Set up a shared GitHub repository with the agreed folder structure.
- Cloud Engineer provisions the AWS account, sets IAM roles, and deploys a hello-world SageMaker endpoint.
- Every teammate reads at least one CalCOFI-related paper for domain grounding.
- Create a shared architecture document with the full system diagram.

8.2 Hackathon Schedule

Hour	Phase	Activities
0 - 4	Kickoff & Setup	Align scope, pull CalCOFI + iNaturalist data, deploy base AWS infra, wireframe surfer dashboard, boilerplate React app.
4 - 12	Data + Baselines	Train naive forecasting baseline, build species occurrence dataset from iNaturalist, label HAB events, build ingestion Lambda, ship static dashboard with mock data. Hour 12 checkpoint: demoable v0.
12 - 20	Real Models	Train PatchTST forecaster, VAE anomaly detector, and species distribution model. Fine-tune ViT on satellite chlorophyll. Deploy SageMaker endpoints, wire dashboard to real APIs.
20 - 28	Integration + LLM	Combine all four models into one inference pipeline, implement the Surf Safety Index fusion layer, add EventBridge and SNS alerting, build Gemini-powered chatbot, integrate species discovery feed.
28 - 34	Polish + Demo Prep	Finalize evaluation tables, write deployment docs, script the 3-minute demo, rehearse. Hour 28 checkpoint: feature freeze.
34 - 36	Submission	Submit to Devpost, record backup demo video, final pitch rehearsal.

9. Scope Discipline: What We Build for Real

In a 36-hour window, not every component can reach production quality. The key to hackathon success is being honest with ourselves about which elements are truly real, which are scoped-down, and which are demonstration-only. This clarity also makes our pitch honest and defensible when judges ask hard questions.

Component	Status	Approach
State Forecaster	Real	Trained for real on a scoped subset of CalCOFI (post-1993 CTD era).
Anomaly Detector	Real	Trained on the same cleaned data as the forecaster.
HAB Classifier	Partial	Fine-tuned on a small labeled set; use chlorophyll threshold as proxy label where needed.
Marine Life Predictor	Real	Gradient-boosted species distribution model trained on iNaturalist occurrences fused with oceanographic features.
Surf Safety Index	Real	Rule-based fusion of Models A, B, C and NOAA wave data, calibrated against historical beach closures.
Satellite Pipeline	Real	Live NOAA CoastWatch API integration.
AWS Infrastructure	Real	Fully deployed across all listed services.
Real-time Streaming	Simulated	Kinesis fully wired; stream replayed historical data during the demo.
LLM Agent	Real	Live Gemini API calls grounded in model outputs and a small RAG index.
Alerts to Stakeholders	Demo	Send a real SMS to a team member's phone on stage to prove the loop closes.

The consistent theme is that the infrastructure is fully real while the data volume is deliberately scoped down. Judges care that the system would scale; they do not need it to process petabytes during the demo.

10. Prize Stack and Submission Strategy

ReefPulse is deliberately designed to be eligible for multiple overlapping challenges. Each submission emphasizes a different facet of the same underlying system. The targeted prize pool exceeds $12,000 across the categories below.

Challenge / Track	Prize Pool	What We Emphasize
ML / AI Track	$5,000	Multi-model fusion (forecaster, anomaly, HAB, species), evaluation rigor.
Cloud Track	$2,000	AWS architecture, event-driven design, MLOps pipeline.
Scripps Challenge	$1,500	Deep CalCOFI usage including larval fish records for species prediction.
Best Use of AWS Services	$1,000	Deep integration across 8+ AWS services.
Most Innovative Build With AI	$1,000	Gemini agent layer and end-user value of the LLM interface.
Best Use of [X] Data	$1,500	Multi-dataset fusion across CalCOFI, gliders, moorings, iNaturalist, and satellite.
Most Viral Idea (SDx)	$300	Surfer- and swimmer-friendly product with a species discovery feed (highly shareable).
Best Use of Gemini API	MLH Swag	Natural-language interface bridging ocean science and the public.

11. Risks and Mitigations

We have identified the following risks and corresponding mitigations upfront. Each has a clear owner on the team.

Risk	Mitigation
CalCOFI historical data is heterogeneous (1949 instruments differ from modern CTDs).	Restrict training to post-1993 CTD-Rosette era for consistency.
Satellite imagery APIs have rate limits during the event.	Pre-download snapshots before the hackathon and cache them in S3.
HAB event labels are sparse and sometimes ambiguous.	Use chlorophyll-a threshold crossings as proxy labels; document the limitation honestly.
iNaturalist sightings are biased toward populated and photogenic areas.	Correct with a presence-only model (MaxEnt-style) and weight by effort; acknowledge the bias in model cards.

Risk	Mitigation
Surf Safety Index could be wrong in ways that endanger users.	Conservative thresholds, clear disclaimers, and always defer to official closure notices from authorities.
SageMaker has a learning curve.	Cloud Engineer pre-deploys a hello-world endpoint before the hackathon begins.
Scope creep during the hackathon.	Enforce hard feature freezes at hour 12 and hour 28.
Live demo failure.	Record a backup demo video at hour 30 as insurance.

12. Resources and References

12.1 Primary Data Sources

- **CalCOFI Bottle Database:** https://calcofi.org/data/oceanographic-data/bottle-database/
- **Spray Glider Data:** https://spraydata.ucsd.edu/data-access
- **CCE Mooring Data:** https://mooring.ucsd.edu/cce/
- **MBARI Data Repositories:** https://www.mbari.org/our-work/data-repositories/
- **iNaturalist Observations:** https://www.inaturalist.org/observations
- **NOAA CoastWatch:** https://coastwatch.pfeg.noaa.gov/
- **NOAA Surf Zone Forecast:** https://www.weather.gov/mfl/surf
- **California Harmful Algal Bloom Reports:** https://mywaterquality.ca.gov/habs/

12.2 Model References

- **PatchTST (time-series transformer):** https://github.com/yuqinie98/PatchTST
- **Hugging Face Vision Transformer:** https://huggingface.co/google/vit-base-patch16-224
- **LightGBM:** https://lightgbm.readthedocs.io/
- **MaxEnt species distribution modeling:** https://biodiversityinformatics.amnh.org/open_source/maxent/

12.3 Cloud and ML Platform

- **SageMaker example notebooks:** https://github.com/aws/amazon-sagemaker-examples
- **AWS architecture diagramming:** https://app.diagrams.net/
- **Gemini API documentation:** https://ai.google.dev/

13. Conclusion

ReefPulse turns seventy-plus years of Scripps oceanographic observations, fused with crowdsourced species sightings and real-time sensor telemetry, into a modern ocean intelligence platform built for the people who actually use the California coast. It unites four machine learning models, a Surf Safety Index fusion layer, a full AWS event-driven architecture, and a natural-language interface into a coherent product that serves surfers, swimmers, fishers, divers, and coastal managers. The marine life discovery feature turns what would otherwise be a pure safety tool into a product users open every day - a viral hook with a serious backbone.

The proposal is intentionally scoped: we will build the infrastructure and the core models for real, we will scope down the data volume, and we will demonstrate the full closed loop during the final pitch. We believe this project represents the strongest possible combination of the Machine Learning / AI track, the Cloud track, and the Scripps Challenge, while also positioning

us for the Best Use of AWS Services, Build with AI, Gemini API, Best Use of [X] Data, and Most Viral Idea challenges. The target prize pool is significant, but more importantly, the project closes a genuine gap between ocean science and the communities that depend on it. With disciplined scope management and the hour-by-hour timeline above, we believe ReefPulse is both ambitious and achievable within the DataHacks 2026 hackathon window.